CASE STUDY · APPLIED AI ENGINEERING

DocQA

Document answers you can audit

Answers questions about a PDF or text document. Every claim cites a page and a quote, checked against the source before anyone sees it.

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Solo build: design, backend, LLM, UI, deploy
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github.com/alizaimoiz6-collab/docqa
Live demo ↗
Scan for live demo

100%	18 / 18	3.5 s	$0.005
of citations verified word-for-word (31 / 31)	test questions handled correctly, incl. 4 correctly declined	median end-to-end answer time on the live app	average cost per question; caching cut it 55%

THE PROBLEM

Teams that work from long documents, such as insurance policies, claims manuals and contracts, spend a long time finding one clause. Generic AI chat gives fluent answers **nobody can check**, and in regulated work an answer without a source can't be used.

THE SOLUTION

A service that returns a **structured answer with page-level citations** and verifies every quote against the document on the server. A fabricated citation is flagged before it reaches the user. When the document doesn't contain the answer, it **says so instead of guessing**.

HOW IT WORKS

01 Upload
PDF or text is parsed into numbered passages that keep their page numbers.

→

02 Ask Claude
The whole document, or its most relevant passages if it's very large, goes to Claude Opus 5.5.

→

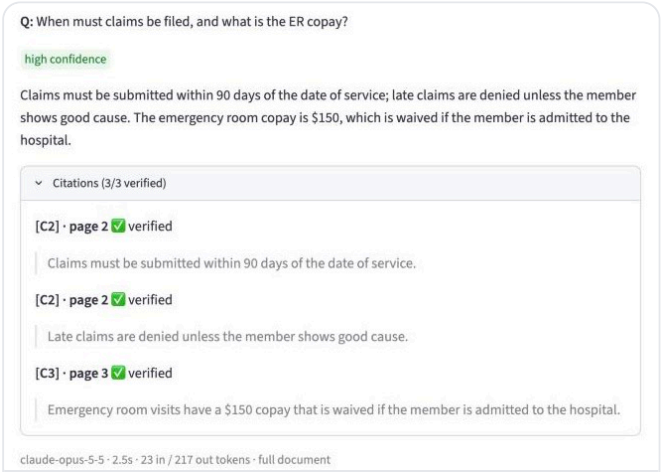
03 Structured answer
Output must match a Pydantic schema: answer, confidence, and passage + quote citations.

→

04 Verify
Each quote is matched against the passage it cites and marked verified or flagged.

KEY DECISIONS

- **Structured outputs, not free text.** Every response parses into a typed object; no fragile text parsing.
- **Server-side citation checks.** Turns "trust the model" into a check that can be measured and audited.
- **Long context first, no vector database.** Fewer missed answers, less infrastructure; a light ranker handles huge files.
- **Prompt caching.** Follow-ups reuse the cached document, cutting cost 55% in testing.
- **Built for failure.** Retries, clear errors for timeouts, per-user rate limits, and a shared $1/day spend cap.



Live app: one answer, three citations, each verified against pages 2–3 of the policy PDF.

BUSINESS VALUE

- **Trust:** reviewers check one quoted line, not a whole file.
- **Speed:** minutes of searching become seconds.
- **Low cost:** serverless, scales to zero when idle.

ENGINEERING QUALITY AND NEXT STEPS

- 32 automated tests plus a reproducible live evaluation.
- Public API with OpenAPI docs; deployed on Modal.
- **Next:** OCR for scanned PDFs, multi-document Q&A, streaming.

Python · FastAPI · Claude API · Opus 5.5 · Pydantic · Streamlit · Modal

Metrics: 18-question test set on three short sample documents (eval/run_eval.py).